

DISCONNECTED COUNTEREXAMPLES TO A REAL-ROOTEDNESS CONJECTURE FOR WEIGHTED BOND POSETS

ABSTRACT. Conjecture 4.13(2) of González D'León and Wachs [1] asserts that a signed difference of weighted-bond-poset Möbius polynomials is real-rooted whenever the underlying graphs have the same numbers of vertices and connected components and one is a subgraph of the other. We show that the conjecture is false for disconnected graphs. For every integer $r \geq 3$, the pair $G_r = rK_3$ and $H_r = rP_3$ is a counterexample, where P_3 is the three-vertex path and rX denotes the disjoint union of r copies of X . The connected restriction is not addressed.

Let $\mu_G(t)$ denote the Möbius polynomial of the weighted bond poset of a graph G , as defined in [1]. Conjecture 4.13(2) of [1] states that if G has n vertices and k connected components and H is a subgraph of G with the same n and k , then

$$(-1)^{n-k}(\mu_G(t) - \mu_H(t))$$

is real-rooted. The conjecture is stated for arbitrary k .

Theorem 1. *For every integer $r \geq 3$, let*

$$G_r = rK_3, \quad H_r = rP_3,$$

where each copy of P_3 is obtained by deleting one edge from the corresponding copy of K_3 . Then H_r is a spanning subgraph of G_r , both graphs have $3r$ vertices and r connected components, and

$$(-1)^{3r-r}(\mu_{G_r}(t) - \mu_{H_r}(t))$$

is not real-rooted. Consequently, Conjecture 4.13(2) of [1] is false as stated.

Proof. Set

$$A(t) = 2 + 5t + 2t^2, \quad B(t) = 1 + 3t + t^2.$$

Equation (1.5) and Proposition 2.2 of [1] give

$$\mu_{G_r}(t) - \mu_{H_r}(t) = A(t)^r - B(t)^r.$$

Since $3r - r = 2r$, the sign in the conjecture is positive.

Let $\zeta = e^{2\pi i/r}$ and $c = \cos(2\pi/r)$. Since $r \geq 3$, ζ is nonreal and $|c| < 1$. The factorization

$$A(t)^r - B(t)^r = \prod_{j=0}^{r-1} (A(t) - \zeta^j B(t))$$

1

shows that the real polynomial

$$\begin{aligned} Q_r(t) &= (A(t) - \zeta B(t))(A(t) - \bar{\zeta} B(t)) \\ &= A(t)^2 - 2cA(t)B(t) + B(t)^2 \end{aligned}$$

divides $A(t)^r - B(t)^r$. For real t ,

$$Q_r(t) = (A(t) - cB(t))^2 + (1 - c^2)B(t)^2.$$

Thus a real zero of Q_r would be a common zero of A and B . But

$$A(t) - B(t) = (1 + t)^2, \quad A(-1) = B(-1) = -1,$$

so A and B have no common real zero. Hence $Q_r(t) > 0$ for every real t . Its leading coefficient is

$$|2 - \zeta|^2 = 5 - 4c > 0,$$

so Q_r is a quartic polynomial with no real zeros. Since Q_r divides $A^r - B^r$, the latter is not real-rooted. $\square$

The first member of the family occurs at $r = 3$. It has nine vertices and three connected components, and

$$\mu_{3K_3}(t) - \mu_{3P_3}(t) = (1 + t)^2(7 + 37t + 63t^2 + 37t^3 + 7t^4),$$

where the quartic factor is strictly positive on $\mathbb{R}$.

Remark 2. The threshold $r \geq 3$ is exact within this family. Indeed,

$$A - B = (1 + t)^2, \quad A^2 - B^2 = (1 + t)^2(3 + 8t + 3t^2),$$

and the quadratic factor has discriminant 28.

Remark 3. These examples do not refute the companion γ -positivity statement, Conjecture 4.12(2) of [1]. Writing $X = (1 + t)^2$, one has $A = 2X + t$ and $B = X + t$, and therefore

$$A^r - B^r = \sum_{j=0}^{r-1} \binom{r}{j} (2^{r-j} - 1)t^j(1 + t)^{2r-2j}.$$

Thus the difference is γ -positive for every $r \geq 1$. The construction also makes no assertion about Conjecture 4.13(1) or the restriction of Conjecture 4.13(2) to connected graphs.

Remark 4. Apart from three quoted inputs the argument above is self-contained, and we isolate them here. Two are the Möbius polynomials $\mu_{K_3}(t) = 2 + 5t + 2t^2$ and $\mu_{P_3}(t) = 1 + 3t + t^2$ of the weighted bond posets of K_3 and P_3 , and the multiplicativity $\mu_{rX} = \mu_X^r$ over disjoint unions; these are Equation (1.5) and Proposition 2.2 of [1], and it is they that identify the polynomials A and B of the proof with μ_{K_3} and μ_{P_3} . The third is the reading of Conjecture 4.13(2) recorded above: it is asserted there for arbitrary k , with no connectivity hypothesis, which is what makes a disconnected pair admissible. Everything else — the identities $A - B = (1 + t)^2$, $A^2 - B^2 = (1 + t)^2(3 + 8t + 3t^2)$ and $A^3 - B^3 = (1 + t)^2(7 + 37t + 63t^2 + 37t^3 + 7t^4)$,

the positivity of Q_r , and the γ -expansion — is polynomial algebra over $\mathbb{Z}$ which the reader can carry out directly.

REFERENCES

- [1] R. S. González D'León and M. L. Wachs, *Weighted bond posets and a new chromatic symmetric function*, arXiv:2608.08692v1 [math.CO] (2026).